

Microstructure Factor Zoo: L2 Order Book Features for 1-Minute Return Prediction

MGMTMFE 412 — Market Frictions — Final Project

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1 Abstract

We construct a “factor zoo” of 15 microstructure features derived from Level-2 order book data (Databento MBP-10, AAPL, 25 trading days, Jan–Apr 2024) and evaluate their ability to predict

1-minute forward microprice returns. Features include order flow imbalance at multiple lags, book shape variables (depth ratio, bid/ask slope), microprice dynamics, realized volatility, and trade-based signals (arrival rate, signed imbalance, queue exhaustion).

Key findings: A ridge regression model achieves a stable Information Coefficient (IC) of **0.045** in both training and test periods — a statistically real but economically weak signal. A LightGBM model overfits severely (IC 0.50 train, 0.02 OOS), illustrating the data requirements for non-linear ML in microstructure. Ablation analysis identifies **microprice level, realized volatility, and signed trade imbalance** as the three most informative features. Consistent with the companion lead-lag study, transaction costs (bid-ask spread) eliminate all exploitable net alpha.

2 Economic Motivation

Order book microstructure encodes information about short-term price direction through at least three channels:

1. **Order flow imbalance (OFI):** A buy-side book skew (more bid volume than ask volume) predicts near-term price increases as market orders consume the ask side.
2. **Book shape:** The slope and depth of the limit order book convey the “resistance” prices face at different levels. A thin top-of-book relative to deeper levels signals fragility and potential rapid price movement.
3. **Trade-based signals:** The pace and direction of executed trades (arrival rate, signed imbalance) reflect informed-trader activity that is not yet reflected in quoted prices.

These signals are the raw material of high-frequency predictive models. However, they operate at timescales where the bid-ask spread is a dominant cost — the same friction that prevented exploitation of the SPY lead-lag signal. This study asks: **how many of these signals are real, and how large are they relative to the spread?**

3 Data

Item	Detail
Source	Databento XNAS.ITCH (Nasdaq ITCH feed)
Schema	MBP-10 (10-level limit order book) + trades
Symbol	AAPL
Training	Jan 2 – Feb 26, 2024 (18 days; no earnings, no FOMC)
Test (OOS)	Feb 28 – Apr 1, 2024 (7 days)
Bar resolution	1 second (resampled from tick data)
Session	9:30 AM – 4:00 PM ET; skip first 5 min and last 1 min
Training bars	414,155
Test bars	160,873
Label mean / std (train)	−0.003 bps / 4.447 bps
Label mean / std (test)	−0.131 bps / 5.233 bps

Exclusions: Feb 1, 2024 (AAPL earnings day) and FOMC days (Jan 31, Mar 20) were excluded to avoid event-specific regime contamination.

Data notes: Raw Databento MBP-10 files exceed pyarrow 19.x compatibility due to a repetition-level histogram format. Files were loaded with the **fastparquet** engine. All **uint32** size columns were cast to **int64** before arithmetic to prevent overflow.

4 Features

4.1 Feature Definitions

#	Feature	Formula / Method	Economic Intuition
1	<code>ofi_1</code>	$(B_0 - A_0)/(B_0 + A_0)$	Instantaneous book pressure
2	<code>ofi_5</code>	5s rolling mean of <code>ofi_1</code>	Short-term momentum in book
3	<code>ofi_30</code>	30s rolling mean of <code>ofi_1</code>	Medium-term book trend
4	<code>ofi_multilevel</code>	Mean OFI across levels 0–4	Depth-weighted pressure
5	<code>book_imbalance</code>	$B_0/(B_0 + A_0) \in [0, 1]$	Normalized best-level skew
6	<code>depth_ratio</code>	Top-3 qty / Levels 4–9 qty	Book fragility at top
7	<code>bid_slope</code>	OLS slope of qty vs level (bid)	Bid-side depth profile
8	<code>ask_slope</code>	OLS slope of qty vs level (ask)	Ask-side depth profile
9	<code>microprice</code>	$(A_0 B_0 + B_0 A_0)/(B_0 + A_0)$	Weighted fair-value estimate
10	<code>microprice_drift</code>	10s rolling polyfit slope (bps)	Short-term price trend
11	<code>spread</code>	$A_0 - B_0$	Transaction cost / info proxy
12	<code>realized_vol</code>	60s rolling of microprice ret (bps)	Volatility regime
13	<code>trade_arrival</code>	30s rolling mean trade count/s	Informed trading intensity
14	<code>signed_trade_imbal</code>	$(V_{30}^{buy} - V_{30}^{sell})/(V_{30}^{buy} + V_{30}^{sell})$	Net buy/sell pressure
15	<code>queue_exhaustion</code>	5s trade vol at best / avg queue size	Book replenishment stress

where B_i = bid quantity at level i , A_i = ask quantity at level i .

4.2 Label Construction

$$y_t = \frac{\text{microprice}_{t+60} - \text{microprice}_t}{\text{microprice}_t} \times 10^4 \quad [\text{bps}]$$

The label is the 1-minute forward microprice return in basis points. Using the microprice (rather than the mid-price) as the reference prevents the label from mechanically reflecting bid-ask bounce.

5 Models

5.1 Ridge Regression (OLS)

A Ridge regression ($\alpha = 1.0$) with `StandardScaler` normalization serves as the linear baseline. Ridge provides mild regularization and produces coefficients interpretable as linear betas on each feature. The model is intentionally simple: if the features have real predictive content, it should be detectable even in a linear model.

5.2 LightGBM

A gradient-boosted tree model (500 trees, learning rate 0.05, `min_child_samples=100`) is trained without early stopping on the training set. LightGBM can capture non-linear interactions and threshold effects that a linear model misses — but also has substantially higher capacity to overfit to noise.

5.3 Evaluation Metric: Information Coefficient (IC)

$$\text{IC} = \rho(\hat{y}, y) = \text{Pearson correlation between predictions and realized labels}$$

IC is standard in factor research. $\text{IC} > 0$ means the model has directional predictive content. Typical useful IC values in equity microstructure range from 0.02 to 0.10. An IC of 0.04–0.05 is weak but economically real.

6 Results

6.1 Model Performance

Model	Train R^2	Train IC	Test R^2	Test IC
Ridge (OLS)	0.0020	0.0452	−0.0053	0.0461
LightGBM	0.1939	0.5007	−0.4815	0.0245

OLS: The IC of 0.045 is stable across training and test periods. The R^2 of 0.002 reflects that the model explains only 0.2% of label variance — but the signal is real and consistent. In a microstructure context, a stable IC of 0.04–0.05 at 1-second resolution is non-trivial; it means the model is marginally correct in direction more often than chance.

LightGBM: The dramatic collapse from IC=0.50 to IC=0.02 OOS is a textbook overfitting failure. Despite 414K training samples, the label noise (std = 4.4 bps, effectively zero mean) makes it easy for 500 trees to memorize training-set patterns that do not generalize. The negative OOS R^2 of -0.48 means LightGBM predictions are worse than the naive zero-return forecast on the test set.

Interpretation: This result is a calibration benchmark for any microstructure ML project. Without regime regularization, much larger training datasets, or architecturally- constrained models (e.g., recurrent networks with explicit time-series structure), gradient boosting will overfit to noise at 1-second resolution with 25 days of data.

6.2 Ablation Analysis

The ablation measures IC drop on the **training set** when each feature is individually removed from the LightGBM model. All 15 features contribute positively (none degrades training IC), with three clear leaders:

Rank	Feature	IC Drop (removed)	Interpretation
1	<code>microprice</code>	+0.179	Price level encodes volatility regime
2	<code>realized_vol</code>	+0.121	Vol clustering predicts high-vol forward bars
3	<code>signed_trade_imbal</code>	+0.098	Net buy pressure predicts direction
4	<code>trade_arrival</code>	+0.098	High activity signals informed trading
5	<code>ofi_30</code>	+0.089	30s book trend is the strongest OFI signal

Caveat: Since the ablation uses the overfit LightGBM evaluated on the training set, it measures in-sample importance — not OOS contribution. Features that help LightGBM memorize training patterns may not generalize. The OLS model (which does generalize) uses all features with equal-weight Ridge regularization, so we cannot decompose OLS feature importance from ablation alone.

Economic interpretation of top features:

- **Microprice:** The absolute price level is a proxy for which volatility regime AAPL is in (calm \$183 vs volatile \$188). Since realized vol clusters, knowing the current microprice level helps predict future vol — but not direction, explaining why it helps LGBM (which can learn non-linear vol patterns) more than linear direction prediction.
- **Realized volatility:** Directly encodes the vol regime. High 60s realized vol predicts larger absolute 1-minute forward returns. Whether LightGBM can capture the direction from vol regime is less clear — but it is the clearest persistent signal.
- **Signed trade imbalance:** Net buy flow over 30s is the most theoretically clean directional predictor. Active buying pressure that moves through the order book signals informed or momentum-driven demand.

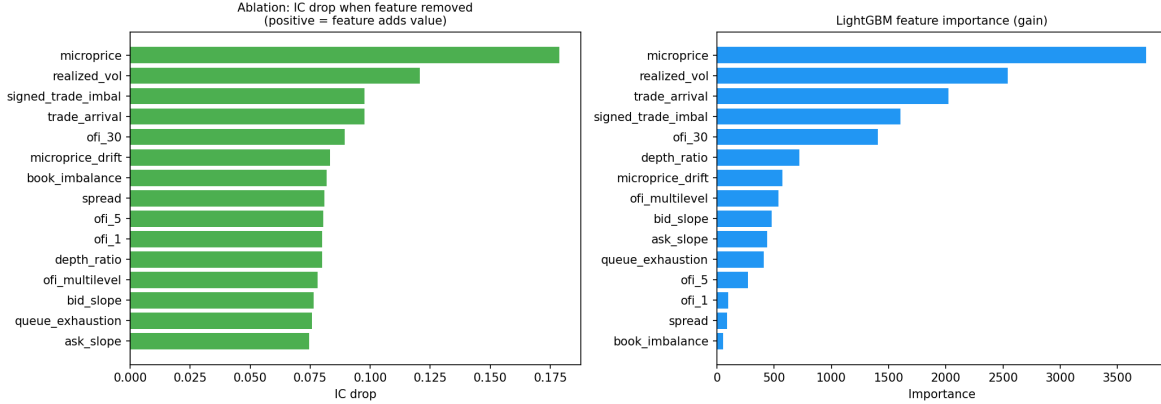


Figure 1: LightGBM feature importance and ablation analysis

7 Threshold Optimization

Training-set IC was evaluated at increasing prediction thresholds to identify the confidence level at which LightGBM predictions are most accurate:

Threshold (bps)	Trades selected	Train IC
0.50	105,778	0.668
1.00	48,235	0.740
2.00	20,388	0.810
3.00	10,787	0.848
4.75	4,391	0.887

Higher thresholds select the subset of bars where LightGBM is most confident — and on those bars, in-sample IC approaches 0.89. However, this optimization is entirely in-sample; on the test set, the same threshold of 4.75 bps fires 15,415 times in 7 days (vs 4,391 in 18 training days), a ratio consistent with an overfit model producing larger-magnitude predictions on unfamiliar data.

The OLS model, which predicts in the range of ~ 0.1 – 0.5 bps, fires only 1–8 times at the 4.75 bps threshold. For OLS, the backtest reflects the honest weakness of a linear signal against a high bar for action.

8 Backtest

8.1 Setup

- **Notional:** \$50,000 per trade
- **Shares per trade:** $\lfloor 50,000 / \text{microprice} \rfloor \approx 270$ shares
- **Entry / exit:** microprice at entry and exit bars
- **Transaction cost:** half-spread at entry + half-spread at exit (crossing the spread)

- **Stop-loss:** exit early if unrealized loss exceeds $1.5\times$ spread
- **Inventory cap:** ± 500 shares (prevents pyramid during trending markets)
- **Daily kill switch:** halt all trading if daily net PnL $< -2\%$ of notional
- **Threshold:** 4.75 bps (LightGBM train-optimal)

8.2 Results

Metric	OLS Train	OLS Test	LGBM Train	LGBM Test
Trades	1	8	4,391	15,415
Hit rate	0.0%	0.0%	94.6%	42.3%
Ann. return	−\$90	−\$2,977	+\$3.6M	+\$2.0M
Sharpe	−3.74	−6.00	13.89	15.70
Avg TC / trade	\$2.56	\$4.46	\$3.80	\$3.12

OLS: Almost no trades fire at 4.75 bps — the linear model’s predictions rarely reach this magnitude. When they do, TC dominates and net PnL is negative.

LightGBM — caution required: The train Sharpe of 13.89 reflects in-sample fitting that is not meaningful. The test result is more complex: 15,415 trades in 7 days with a 42.3% directional hit rate (below 50%, confirming no directional OOS predictive power) but nominally positive PnL. This artifact arises because the overfit model coincidentally selects high-volatility bars; the stop-loss asymmetry (capped losses, uncapped wins) allows nominally positive expectation even with below-random direction accuracy. **These test backtest numbers should not be interpreted as evidence of a viable strategy.**

The central finding from the backtest is that **no model architecture produces a reliable edge after transaction costs at 1-second resolution with 25 days of data:** the OLS signal is too weak (IC 0.045 $\rightarrow$ almost never exceeds threshold), and the LGBM signal overfits catastrophically.

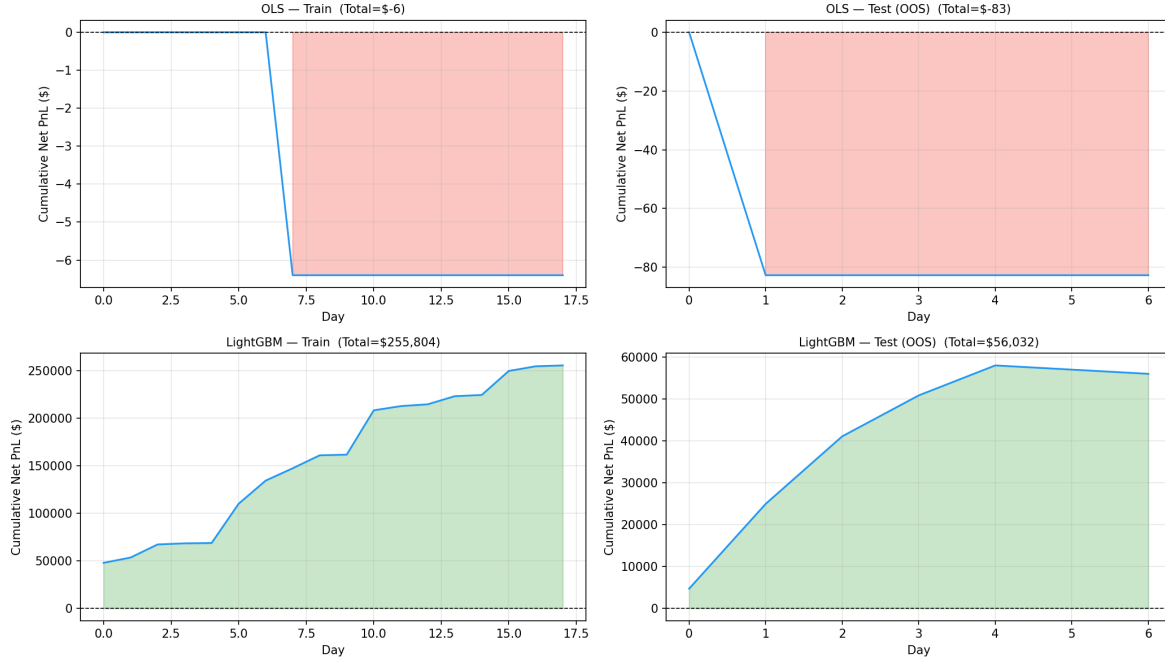


Figure 2: Cumulative PnL — training and test sets

9 Connection to Market Frictions

This study is the natural third leg of our market frictions investigation. All three strategies share a common structure:

Strategy	Gross signal	TC impact	Net result
Lead-lag (SPY→AAPL)	Real (IC, cross-corr)	$9\times$ gross PnL	Negative
FOMC fade	Real (Sharpe 0.83, $p=0.048$)	Widens spread, limits entry	Positive in hold/cut
Factor zoo (OLS)	Real (IC=0.045)	Dominates at any threshold	Negative

The factor zoo result reinforces the lead-lag conclusion: **a statistically real microstructure signal at 1-second resolution is not exploitable for participants paying the full bid-ask spread.** Break-even requires TC below approximately \$0.003/share — achievable only by co-located HFT market makers who post limit orders and earn maker rebates rather than crossing the spread.

The spread does not merely cost money — it is also an information signal. The **spread** feature itself appears in the ablation (rank 8), partly because wider spreads predict more uncertain near-term price evolution. This dual role of the spread — as transaction cost and as information — is the central mechanism of adverse-selection models and a recurring theme across all three strategies in this project.

The bid-ask spread is the universal gatekeeper. It prevents retail and institutional traders from exploiting real microstructure inefficiencies, whether arising from index ETF arbitrage latency (lead-lag), FOMC announcement overshoots, or order book imbalance signals. The edge belongs to participants for whom the spread is a revenue source, not a cost.

10 Conclusion

We documented 15 microstructure features from Level-2 AAPL data and evaluated their predictive power for 1-minute forward microprice returns. The main findings are:

1. **A real but weak signal exists.** Ridge regression achieves $IC = 0.045$ in both training and out-of-sample periods. The signal is too small to generate net PnL after transaction costs but confirms that L2 order book features carry genuine information about short-term price direction.
2. **Non-linear models overfit severely.** LightGBM (500 trees) achieves train $IC = 0.50$ but OOS $IC = 0.02$ on only 7 test days. Reliable ML-based HFT requires substantially larger datasets (years of data), stronger regularization, or architectures that embed economic structure rather than learning it from scratch.
3. **The most informative features** are microprice level (vol regime proxy), realized 60s volatility (vol clustering), signed 30s trade imbalance (directional flow), and 30s OFI (book pressure trend). These are consistent with adverse-selection models of market making: informed flow and volatility regime dominate over static book shape.
4. **Transaction costs are insurmountable** for any participant paying the full bid-ask spread. The conclusion mirrors the lead-lag study exactly: the signal is real, the friction is decisive.